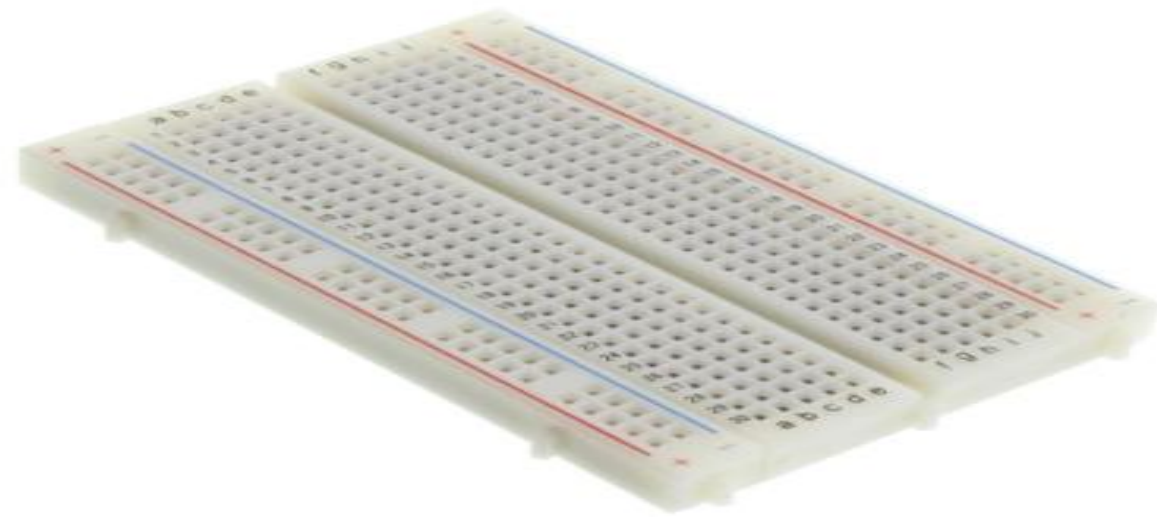
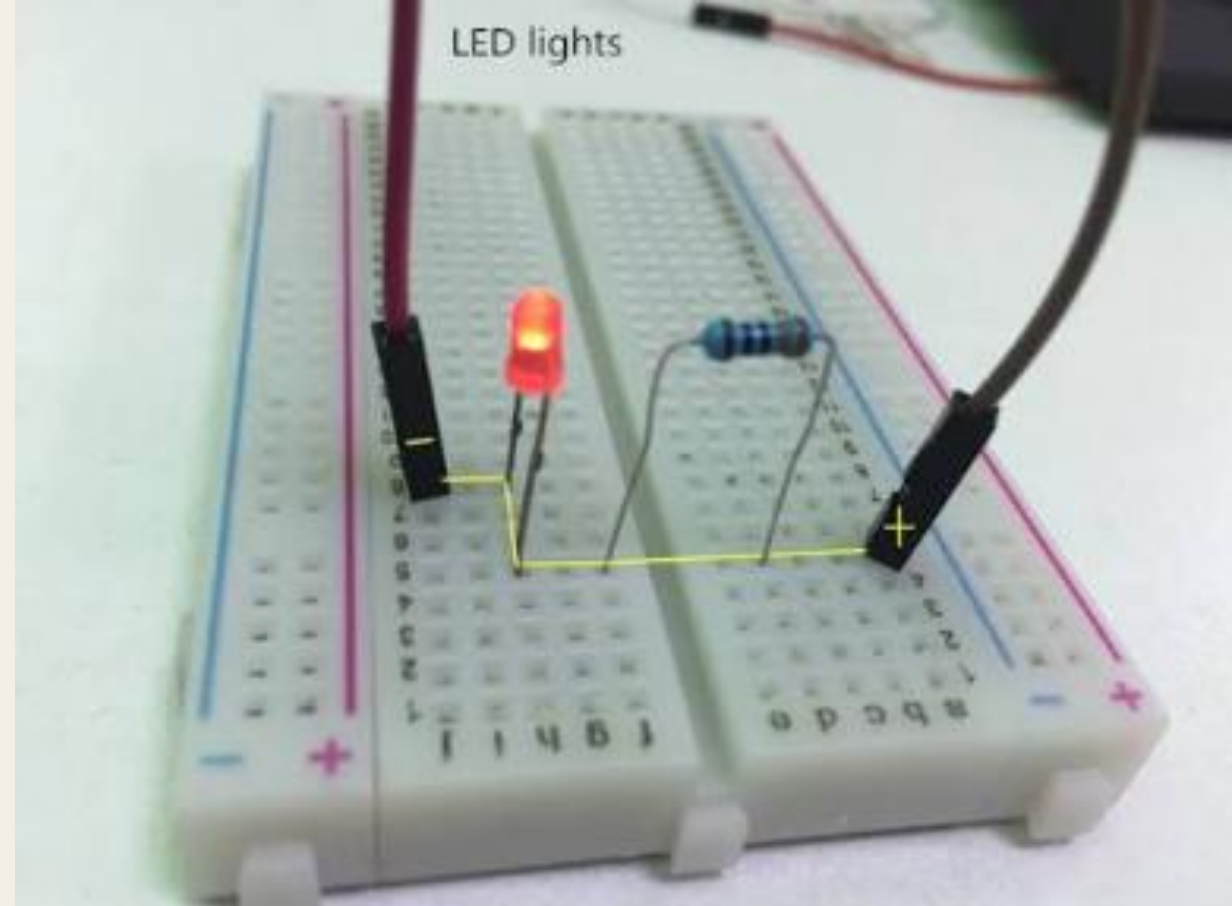


LESSON 04

Breadboard & Circuit Building

Build, test, and rearrange a circuit without soldering.



A breadboard turns circuit ideas into quick experiments

<https://youtu.be/W6mixXsn-Vc?si=19hOvWgFo8D-igur>

It is a temporary circuit-building surface. Parts plug into holes that connect internally, so you can move, test, and rearrange them without soldering or damaging them.



Plug in

Components press into the holes.



Test

Power the idea and observe what happens.



Move

Change one connection at a time.

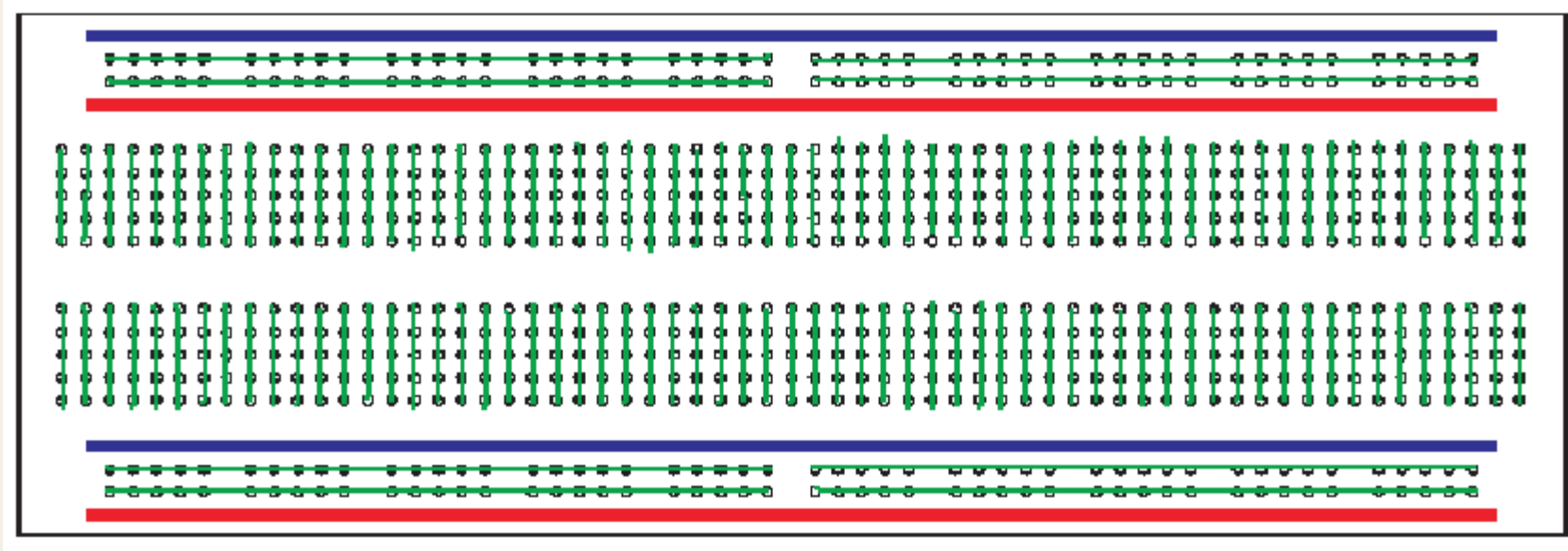


Practise

Start with LEDs, switches, resistors, and controllers.

The center gap separates two connected halves

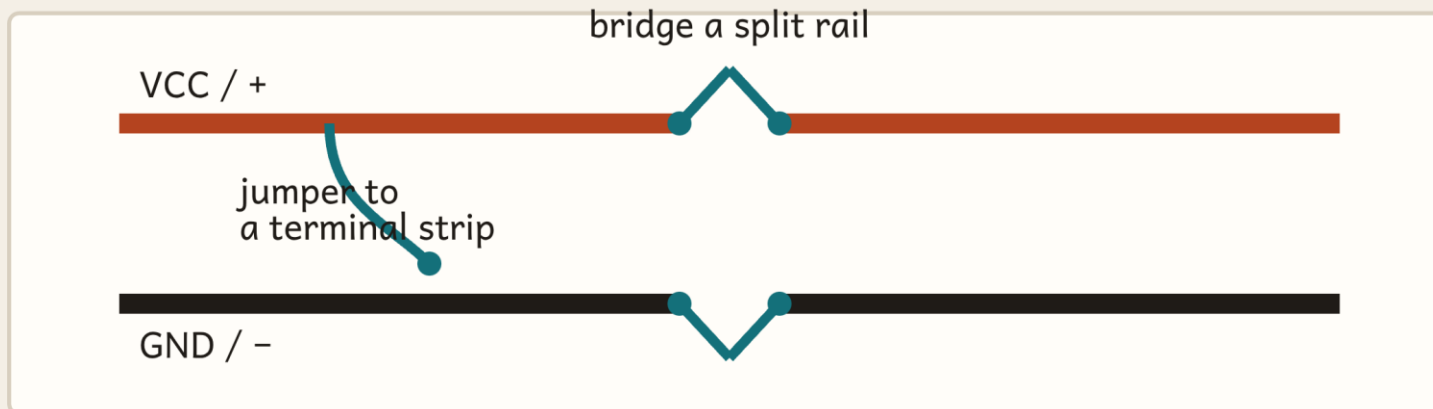
Terminal strips run in connected groups on each side of the center gap. Holes in one five-hole group are electrically connected; holes across the gap are not. Power rails run along the edges and distribute VCC/+ and GND/-.



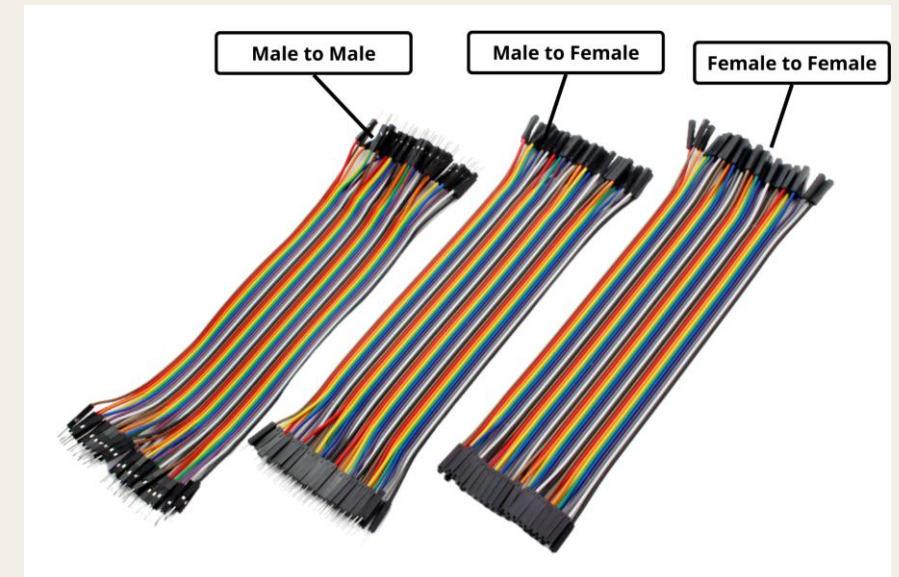
Power rails make VCC and GND easy to distribute

Use the red or + rail for VCC/+ and the blue, black, or – rail for GND/–. Run jumpers from a rail to any terminal strip that needs power.

BREADBOARD EDGE



Some boards break each rail in the middle.



Jumper wires are the temporary connections between parts

<https://youtu.be/pJXCYjq2TQ?si=l8E06Zx2FSMG3HA8>

Choose a wire long enough to route cleanly. Press each end firmly into place.



Male-to-male

Pin to pin. Common for breadboard-to-breadboard links.



Male-to-female

Breadboard pin to a controller, sensor, or module header.

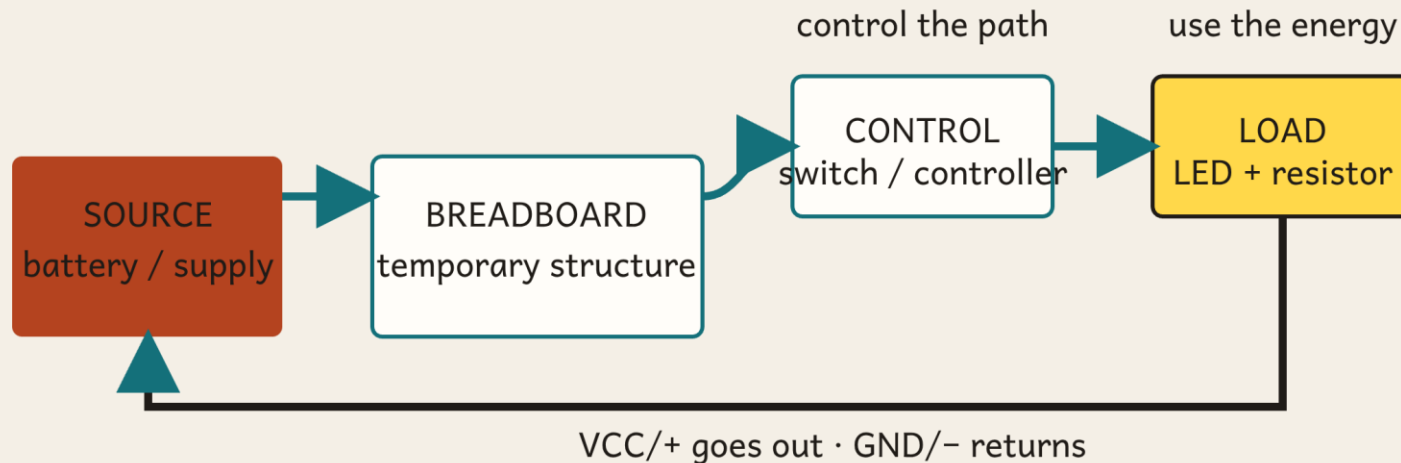


Female-to-female

Header-to-header connections between modules.

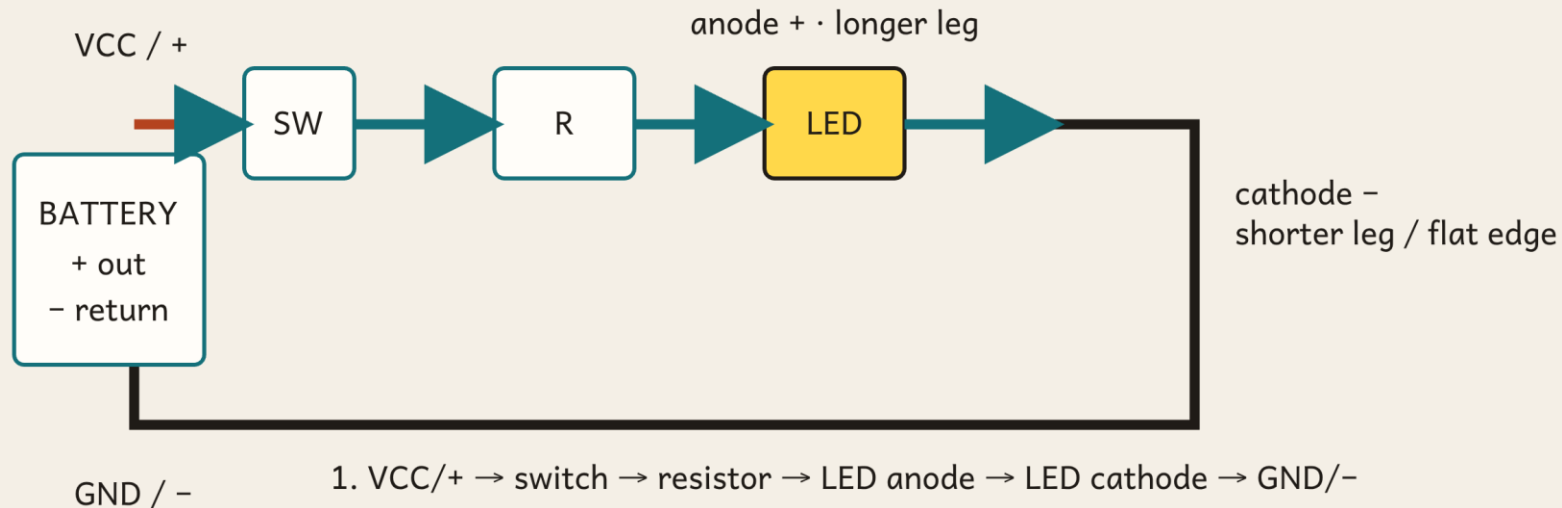
A basic circuit needs a source, control, and load

Energy starts at the source, travels through a controlled path, and reaches the load. The breadboard gives every part a temporary place to connect.



Example: press the switch to light the LED

Build the path in order. The LED lights only when the circuit is closed through the intended route.



Press closes the path.

Release opens the path.

Common mistakes to catch before powering up

Power off, trace the path, then test again.

- ☐ **Wrong rail connection** — check that VCC/+ and GND/- are on the intended rails.

- ☐ **Short circuit** — never connect positive directly to ground; inspect the path first.

- ☐ **Loose jumper** — reseal both ends if the circuit flickers or does nothing.

- ☐ **Reversed LED polarity** — anode faces the resistor; cathode returns to GND/-.